Lab 7: Indicator Variables and Multicollinearity

ACMS 30600 - Fall 2025

Lab Objectives: by the end of this lab you should be able to

- Create and interpret a linear regression model with one categorical predictor.
- Create and interpret a linear regression model with two categorical predictors.
- Use the context of a problem to determine if there might be multicollinearity present in a linear model.
- Compute variance inflation factors (VIFs) for predictor variables of a linear model to determine if multicollinearity is present in the model.
- Use the `stepAIC()` function in R for model selection.

Part 1: Categorical Predictor Variables

The file *gavote.txt* contains data on 159 counties in Georgia for the 2000 US presidential election. There are 10 variables, but we will mainly be interested in: `econ`, a qualitative variable with levels poor, middle, and rich; `rural`: whether the county is considered rural or urban; `votes`, the number of votes recorded in a given county; and `ballots`, the number of ballots issued in a given county.

A ballot is issued if a person goes to a polling station and successfully shows that he/she is registered to vote. Thus, if the value for ballots is greater than votes for a given county that means a certain number of votes were not recorded for people who were issued ballots. This may be due to any of a number of reasons such as a ballot being invalid because the person voted for more than one candidate, the equipment fails to record the vote, etc. The quantity (ballots-votes) is referred to as the "undercount". *We will view the undercount as a proportion, i.e., the proportion of total ballots issued in a county that were undercounted*.

Load the *gavote.txt* data into R. Name the dataset `gavote`.

1. (1 point) Calculate the `undercount` for each observation. That is, find the proportion of total ballots issued in a county that were undercounted. What is the code to do this?
2. (2 points) Fit a linear regression model using `econ` as the predictor of `undercount`. Look at the summary of the model.
    a. Which level of `econ` did R use as the reference (or baseline) level?
    b. Why did R use this level?
3. (2 points) What are the value and interpretation of $\hat{\beta}_{econ:poor}$?
4. (2 points) What is the predicted undercount for a county with `econ` taking the value `middle`? Explain how you know.
5. (3 points) Fit a linear regression model that uses both `econ` and `rural` as predictor variables of `undercount`. Look at the summary of the model.
    a. What is the code to find this model?
    b. What is the linear model based on the summary information?

6. (2 points) What are the characteristics of a county that is represented by just the intercept in this model?

Part 2: Multicollinearity

Load the `seatpos` data in the *faraway* package. Read the help file for the data (type in R: `help(seatpos)`), to learn more about the variables in the data set.

Our goal is to fit a linear regression model with `hipcenter` as the response and `Ht`, `HtShoes`, `Thigh` and `Leg` as predictors.

7. (2 points) Based on the description of the variables given in R, do you believe multicollinearity will be present in this model? Justify your response.
8. (2 points) Compute the variance inflation factor (VIF) for `Ht` by hand (meaning, don't use the built in R function to find this value; use the equation discussed in class).

Fit the desired linear regression model described above. Name your model `hip.regress`.

9. (3 points) Use the code `vif(hip.regress)`.
    a. What does the given code do?
    b. What do you conclude about the model?
10. (2 points) Use the `stepAIC()` function in the *MASS* library, with backwards selection. Can any of the variables be removed from the `hip.regress` model?
11. (4 points) Fit the final model recommended by the `stepAIC()` procedure. Call this model `hip.regress1`.
    a. Find the R^2 and Adjusted R^2 for both the `hip.regress` and `hip.regress1` models.
    b. Does it appear that `hip.regress1` has a sizable difference in predictive ability than `hip.regress`?